

ANKARA MURTED, Turkey

WMO: 171270

Lat: 40.079N

Lon: 32.566E

Elev: 843

StdP: 91.59

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-12.0	-9.8	-14.2	1.2	-9.5	-12.2	1.5	-7.2	8.6	1.2	7.6	1.1	1.2	0	0.521

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	15.8	35.0	17.2	33.2	17.0	31.9	16.8	19.2	29.4	18.5	28.8	17.8	28.2	3.7	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.0	12.6	22.5	15.1	11.9	22.0	14.1	11.2	21.2	58.4	29.3	55.8	28.7	53.5	28.3	24.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.7	7.4	6.4	-15.2	37.5	2.7	1.5	-17.2	38.6	-18.8	39.5	-20.3	40.3	-22.3	41.4		
(5)			-16.6	20.6	3.3	1.4	-19.0	21.6	-21.0	22.4	-22.8	23.1	-25.2	24.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	11.6	0.1	2.3	5.6	10.1	15.2	19.5	23.3	23.8	18.5	12.6	5.7	1.8
	DBStd	8.89	4.63	4.34	3.37	3.31	3.12	2.81	2.65	2.43	3.55	3.60	3.50	4.18
	HDD10.0	1109	307	217	140	38	1	0	0	0	17	135	254	
	HDD18.3	2894	566	450	394	248	105	19	1	41	178	380	512	
	CDD10.0	1693	0	1	4	41	163	286	412	428	254	98	6	1
	CDD18.3	434	0	0	0	0	9	54	155	169	46	1	0	0
	CDH23.3	5826	0	0	0	16	232	775	1965	2057	708	73	0	0
	CDH26.7	2522	0	0	0	0	55	273	932	992	263	7	0	0
Wind	WSAvg	2.6	2.0	2.3	2.8	2.8	2.7	2.9	3.5	3.4	2.8	2.2	1.8	1.9
Precipitation	PrecAvg	375	44	34	34	42	47	33	14	12	17	24	29	48
	PrecMax	626	99	90	83	94	89	92	51	43	95	69	117	97
	PrecMin	248	4	7	8	6	17	2	0	1	0	0	2	5
	PrecStd	78	28	23	20	25	19	27	15	11	20	19	23	23
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.0	17.4	20.2	25.8	30.2	34.0	37.8	37.8	34.2	27.2	20.8	14.0
		MCWB	7.1	8.9	9.8	13.0	15.4	16.8	17.5	17.8	15.7	13.7	10.9	9.6
	2%	DB	9.9	14.1	17.8	22.8	28.1	31.8	35.0	35.1	32.0	25.2	17.9	11.2
		MCWB	6.1	7.6	8.8	11.8	15.1	16.8	17.4	17.3	15.5	13.2	10.2	7.9
	5%	DB	8.0	11.9	15.8	20.5	26.0	29.9	33.1	33.8	29.9	23.2	15.1	9.8
		MCWB	5.3	6.6	8.0	11.0	14.5	16.5	17.2	17.2	15.0	12.8	9.1	7.3
	10%	DB	6.1	9.1	13.2	18.2	23.8	28.0	31.8	31.9	27.7	21.0	12.9	8.0
		MCWB	4.5	5.4	7.3	10.0	13.9	16.0	17.0	16.8	14.5	12.2	7.8	6.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	7.8	10.0	10.9	13.7	17.3	19.1	20.2	20.5	17.9	15.2	12.7	10.4
		MCDB	10.4	15.8	16.9	21.8	26.1	28.4	30.4	31.1	27.7	21.5	18.3	12.9
	2%	WB	6.5	8.2	9.5	12.7	16.1	18.2	19.2	19.5	17.0	14.5	11.1	8.6
		MCDB	8.8	12.5	15.7	20.6	24.6	27.4	29.5	29.8	27.2	21.6	15.5	10.7
	5%	WB	5.5	7.0	8.6	11.7	15.4	17.5	18.5	18.6	16.1	13.7	9.6	7.5
		MCDB	7.3	10.9	14.2	18.8	23.6	26.5	29.1	29.1	26.3	21.2	14.3	9.2
	10%	WB	4.5	5.8	7.6	10.8	14.5	16.8	17.8	17.9	15.4	13.1	8.5	6.4
		MCDB	6.1	8.9	12.6	17.2	22.3	25.6	28.6	28.5	25.2	19.7	12.3	7.8
Mean Daily Temperature Range	5% DB	MDBR	8.4	11.3	12.8	14.1	14.6	14.8	15.9	15.8	15.9	14.8	13.6	9.1
		MCDBR	11.7	15.8	18.0	18.2	18.3	17.8	17.5	17.9	19.0	18.6	17.5	11.3
		MCWBR	8.2	9.8	10.1	9.1	7.7	6.2	5.4	5.8	7.1	8.8	10.5	8.0
	5% WB	MCDBR	9.8	13.5	15.0	15.8	16.1	15.6	15.8	15.5	15.8	15.2	14.8	9.9
		MCWBR	7.1	8.6	8.7	8.5	7.3	6.1	5.4	5.3	6.1	7.5	9.3	7.1
Clear-Sky Solar Irradiance	taub	0.303	0.328	0.371	0.407	0.387	0.368	0.365	0.363	0.345	0.344	0.314	0.304	
	taud	2.382	2.332	2.238	2.168	2.291	2.363	2.369	2.353	2.402	2.403	2.456	2.439	
	Ebn at Noon	858	884	877	866	891	906	905	898	896	855	842	830	
	Edh at Noon	84	103	126	144	131	122	120	118	105	95	78	74	
All-Sky Solar Radiation	RadAvg	1.87	2.86	4.00	5.12	6.09	6.92	7.36	6.55	5.15	3.51	2.43	1.66	
	RadStd	0.23	0.42	0.36	0.47	0.46	0.61	0.32	0.29	0.36	0.26	0.25	0.21	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (10 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page